

Stage 5 — Practice and theory

Where the Shortcut Sits: supervisor meeting 5 of 6

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In one sentence. On unaltered test sets masking harms exactly the cases the location law says it should — shortcut-conflicting images with an in-ROI artifact — and on the patient-disjoint thyroid split it lowers sensitivity at every validation-fixed operating point; the crossover replicates in 17 of 18 frozen encoder-cohort pairs, in 3 of 5 fine-tuned networks and on new patients (ISIC 2020, +0.250 [+0.174, +0.327]); a two-parameter theory predicts held-out reversed AUROC with mean error 0.052, less than half the error of the best heuristic.

Notation

$Y \in \{0, 1\}$	diagnosis (1 = melanoma / malignant / solid-component tumour / erosion).
$A \in \{0, 1\}$	artifact present (hair, sonographer caliper, capsule debris, chest tube).
ROI	region of interest (lesion, nodule, tumour, lesion box, lungs); drawn in green ; artifacts in red .
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (in-ROI); Trap B: $r < 0.1$ (out-of-ROI).
Environments	training and correlated test $P(A=1 Y=1) = 0.9$, $P(A=1 Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real traps: half of each class; sweeps: absent).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ of reversed-test AUROC (real traps: Trap B – Trap A; sweeps and transplant: location interaction on the same images).
CI	95% crossed seed×image bootstrap interval (Stage 1, Sec. ??); every interval in these reports uses it.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted.
SMD	standardised mean difference between two groups.

1 Where we are

Stages 2–4 established the location law in constructed environments — a strong artifact–label correlation in training and its reversal at test — and showed that it is not explained by measured confounders. A clinician will ask two questions this cannot answer: does it matter on data as it is collected, without an imposed correlation, and at the operating point a clinic would use? And is the result a property of one encoder, or something that can be predicted? This meeting answers both, reports the breadth of replication (encoder families and sizes, fine-tuning, new patients), and explains the one encoder that does not show the crossover.

2 Question

- Q5a** On natural test sets, does masking harm the shortcut-conflicting cases, and does that change clinical sensitivity at fixed operating points?
- Q5b** How broadly does the crossover replicate?
- Q5c** Can a simple model predict, before seeing it, how a model will perform on the reversed test — and so explain when masking helps?

Design decisions *Hard pairs.* On a natural test set the artifact–label association is weak, so the overall AUROC barely moves. We therefore split positive–negative pairs by whether their artifact status conflicts with the test set’s own association (“hard”) or follows it (“easy”); the law predicts harm on hard pairs. *Operating points fixed on validation data* (never on test) at five clinically motivated targets, with thresholds re-estimated in every bootstrap replicate. *Theory judged by held-out error*, not by fit: the model is fitted to each cell’s clean and correlated AUROC and must predict the reversed AUROC it never saw, against two parameter-free heuristics. Rejected: judging the theory by agreement with simulations drawn from its own assumptions (circular).

3 Methods

Natural test sets Thyroid: the official patient-disjoint TNCD test split (Gong et al., 2022); dermoscopy: each source (BCN, HAM, MSK) held out in turn; capsule: held-out frame blocks. Models are trained on the natural training distribution (no imposed correlation).

Clinical metrics and operating points AUROC, AUPRC, Brier score, expected calibration error; sensitivity and specificity at OP1 (maximal balanced accuracy), OP2/OP3 (validation specificity $\geq 0.80/0.90$) and OP4/OP5 (validation sensitivity $\geq 0.80/0.90$), overall and among shortcut-conflicting positives.

Replication DINOv2 ViT-S/B/L, MedSigLIP-448, ConvNeXt-384 and DermLIP (frozen, linear heads); ResNet-50 and ViT-S fine-tuned end to end; ISIC 2020 as an external dermoscopy cohort with patient-disjoint folds.

Theory A linear-Gaussian model with disease signal-to-noise S and artifact signal-to-noise A , an LDA-optimal head and the trap’s training association predicts the AUROC of any test composition (docs/THEORY.md). Per cell, (S, A) is fitted to the clean and correlated AUROC; the reversed AUROC is predicted. Heuristics: “reversed = clean” and “symmetric reversal” (reversed = clean – (correlated – clean)).

4 Experiments

Table 1: Experiments of this stage. Commit times come from this machine and are not independent proof of order.

Experiment	Status	Registration	First commit (local time)	Support criterion
Natural test sets, N3 (hard pairs)	supported registered	NATURAL	74bd243, 09:39	mask – ERM < 0 on hard pairs
Operating points S1–S3	supported registered	REVIEW2 (R3)	29cd0da, 18:56	“masking lowers sensitivity” only if S1 and ≥ 3 of OP2–OP5
Thyroid subgroup, $n=78$ (S3)	partly supported registered test in a post hoc subgroup	REVIEW2 (R3)	29cd0da, 18:56	sensitivity loss in the subgroup, CI < 0
Crossed CIs for the subgroup (R6)	supported registered	REVIEW3 (R6)	85da735, 23:43	report corrected CI
Encoder scale (S/B/L)	supported registered	SCALE	13303b0, 14:13	crossover CI > 0 at every scale
Fine-tuned networks	supported registered	FINETUNE	7c8b466, 07:06	crossover CI > 0 (secondary)
DermLIP hair traps	supported registered	ISIC2019__TRAPS	76e25d4, 13:14	crossover CI > 0
Theory as held-out predictor, T1–T4	supported registered	THEORY__PREDICTION	4f78653, 13:46	MAE below both heuristics; crossover signs
External validation, X1 (ISIC 2020)	supported registered	FINAL (A4)	c45d482, 00:35	gate, then crossover CI > 0

5 Results

5.1 Natural test sets

Table 2. On all pairs, masking changes little. On shortcut-conflicting pairs it harms in thyroid (-0.095 [$-0.165, -0.022$]), BCN (-0.025 [$-0.043, -0.008$]) and MSK (-0.065 [$-0.111, -0.019$]), and helps in HAM ($+0.050$ [$+0.030, +0.070$]); in capsule the interval includes zero ($+0.021$ [$-0.003, +0.048$]). Where masking harms the hard pairs, it helps or is neutral on the easy ones (MSK $+0.100$ [$+0.056, +0.147$]) — the signature of a model that still uses an in-ROI artifact after masking. The registered claim N3 is thus supported in three of five natural test sets.

Table 2: Natural test sets: AUROC change from masking on all positive–negative pairs and on the pairs whose artifact status conflicts with (“hard”) or follows (“easy”) the test set’s own artifact–label association (DINOv2, crossed CIs).

Test set (no imposed correlation)	$P(A=1 Y=1) / P(A=1 Y=0)$	mask – ERM, all pairs	shortcut-conflicting pairs	shortcut-aligned pairs
Thyroid (official patient-disjoint split)	0.331 / 0.434	-0.004 [$-0.048, +0.041$]	-0.095 [$-0.165, -0.022$]	$+0.042$ [$-0.011, +0.097$]
Dermoscopy, BCN held out	0.356 / 0.239	-0.025 [$-0.035, -0.016$]	-0.025 [$-0.043, -0.008$]	-0.022 [$-0.035, -0.008$]
Dermoscopy, HAM held out	0.274 / 0.184	$+0.031$ [$+0.017, +0.045$]	$+0.050$ [$+0.030, +0.070$]	$+0.004$ [$-0.019, +0.027$]
Dermoscopy, MSK held out	0.109 / 0.074	$+0.016$ [$-0.007, +0.038$]	-0.065 [$-0.111, -0.019$]	$+0.100$ [$+0.056, +0.147$]
Capsule (held-out frames)	0.101 / 0.232	$+0.077$ [$+0.059, +0.097$]	$+0.021$ [$-0.003, +0.048$]	$+0.071$ [$+0.040, +0.108$]

5.2 Clinical consequences: thyroid operating points

On the official, patient-disjoint thyroid test split, masking lowers sensitivity at every operating point (Table 3): at OP1 by -0.275 [$-0.435, -0.138$] and at OP2–OP5 with intervals below zero in 4 of 4. By the decision rule registered before the operating points were computed (S1 and a loss at ≥ 3 of OP2–OP5), the paper may say that masking lowers sensitivity on this test set. Specificity rises: masking moves the operating point, and the loss falls on the malignant nodules whose caliper lies on the nodule.

The subgroup of 78 malignant nodules **Registration status: a pre-registered test (S3) inside a post hoc subgroup — exploratory confirmation.** The subgroup (malignant nodules with an in-ROI caliper) was defined after the natural-distribution AUROC results were known; the operating-point test in it was registered before the operating points were computed. The same 78 nodules (of the official test split) enter every seed. At OP2, sensitivity in this subgroup falls from 0.549 to 0.177, a change of -0.372 $[-0.499, -0.236]$ with the corrected bootstrap. The effect is large and consistent across operating points, but its confirmatory weight is limited by the post hoc subgroup definition; it is not part of the 16-test primary family.

Table 3: Thyroid, official patient-disjoint test split: effect of masking at five validation-fixed operating points (crossed bootstrap with thresholds re-estimated in every replicate). The subgroup is the malignant nodules with an in-ROI caliper (shortcut-conflicting).

Operating point (fixed on validation)	Sensitivity ERM → mask	Δ sensitivity	Δ specificity	Sensitivity, conflicting malignant (ERM → mask)	Δ in that subgroup
OP1 max. balanced accuracy	0.699 → 0.424	-0.275 $[-0.435, -0.138]$	+0.202 $[+0.115, +0.330]$	0.703 → 0.223	-0.479 $[-0.640, -0.313]$
OP2 specificity ≥ 0.80	0.557 → 0.355	-0.202 $[-0.300, -0.100]$	+0.117 $[+0.059, +0.172]$	0.549 → 0.177	-0.372 $[-0.499, -0.236]$
OP3 specificity ≥ 0.90	0.371 → 0.219	-0.152 $[-0.253, -0.043]$	+0.075 $[+0.029, +0.122]$	0.336 → 0.100	-0.236 $[-0.383, -0.089]$
OP4 sensitivity ≥ 0.80	0.753 → 0.380	-0.373 $[-0.469, -0.270]$	+0.278 $[+0.203, +0.359]$	0.744 → 0.197	-0.546 $[-0.671, -0.421]$
OP5 sensitivity ≥ 0.90	0.869 → 0.588	-0.281 $[-0.391, -0.178]$	+0.339 $[+0.236, +0.425]$	0.862 → 0.400	-0.462 $[-0.614, -0.310]$

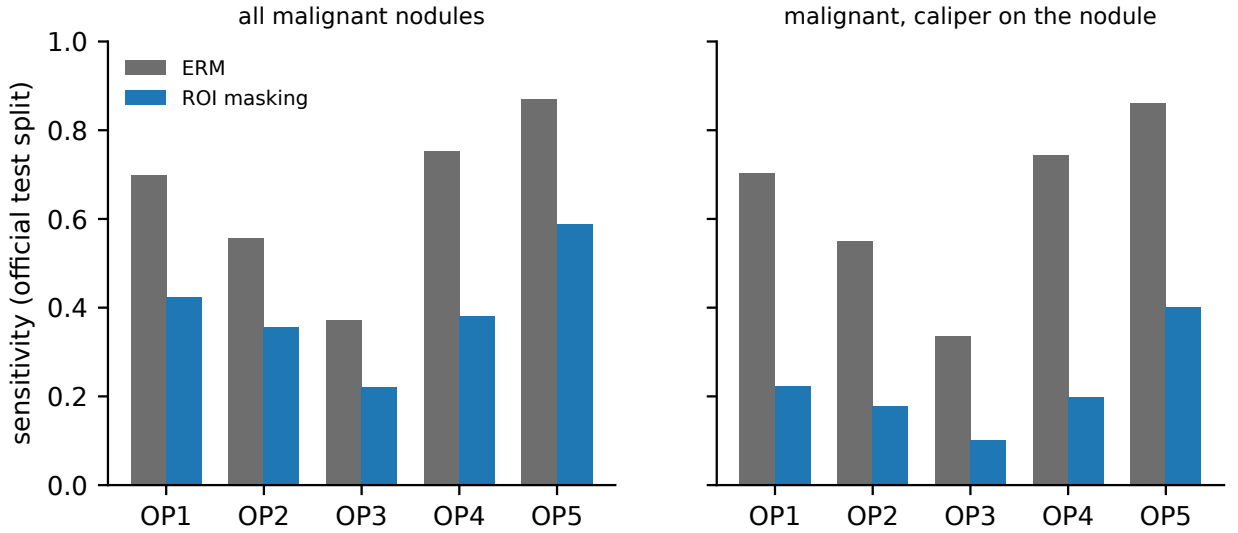


Figure 1: Thyroid, official patient-disjoint test split: sensitivity of ERM and of the masked model at five validation-fixed operating points, for all malignant nodules (left) and for the 78 malignant nodules with a caliper on the nodule (right; post hoc subgroup, exploratory).

Table 4: Other natural test sets at OP2 (validation specificity ≥ 0.80), mask – ERM, crossed CIs.

Test set	Conflicting positives	Δ sensitivity	Δ specificity	Δ sensitivity, conflicting
Dermoscopy BCN	1832	+0.017 $[-0.003, +0.036]$	-0.110 $[-0.135, -0.079]$	+0.019 $[-0.004, +0.042]$
Dermoscopy HAM	808	-0.062 $[-0.110, -0.015]$	+0.092 $[+0.064, +0.121]$	-0.055 $[-0.104, -0.007]$
Dermoscopy MSK	492	-0.074 $[-0.127, -0.026]$	+0.083 $[+0.064, +0.110]$	-0.094 $[-0.149, -0.044]$
Capsule	38	+0.140 $[+0.082, +0.222]$	+0.041 $[-0.029, +0.107]$	+0.022 $[-0.041, +0.123]$

Table 5: Clinical metrics on the natural test sets (seed means; threshold at maximal balanced accuracy on validation data).

Test set	Arm	AUROC	AUPRC	Brier	ECE	Sensitivity	Specificity
Capsule (held-out frames)	ERM	0.889	0.897	0.130	0.042	0.770	0.844
	mask	0.967	0.971	0.069	0.029	0.856	0.933
ISIC, BCN held out	ERM	0.786	0.587	0.238	0.279	0.842	0.521
	mask	0.761	0.564	0.287	0.330	0.842	0.443
ISIC, HAM held out	ERM	0.752	0.355	0.161	0.233	0.575	0.789
	mask	0.782	0.396	0.138	0.199	0.569	0.831
ISIC, MSK held out	ERM	0.770	0.477	0.171	0.160	0.600	0.790
	mask	0.785	0.497	0.149	0.120	0.587	0.823
Thyroid (official split)	ERM	0.735	0.604	0.209	0.088	0.699	0.653
	mask	0.731	0.629	0.215	0.119	0.424	0.856

5.3 Replication breadth

Table 6. The crossover’s interval lies above zero for 17 of 18 frozen encoder-cohort pairs (DINOv2 at three sizes, MedSigLIP, ConvNeXt); the exception is DermLIP. Among end-to-end fine-tuned networks it replicates in 3 of 5 (dermoscopy hair, thyroid ViT-S, capsule); the thyroid ResNet-50 and the ovary ResNet-50 are not significant, and the ovary network barely learns the task, so that null is uninformative. On **new patients** — ISIC 2020, 32,997 images from 2,056 patients, patient-disjoint folds, hair masks from a segmenter trained only on ISIC 2019 and frozen before any ISIC 2020 image was scored — masking harms with hair on the lesion (-0.064 [$-0.128, -0.002$]) and helps with hair beside it ($+0.186$ [$+0.118, +0.249$]); the crossover is $+0.250$ [$+0.174, +0.327$] (registered claim X1, supported).

Table 6: Replication breadth of the real-artifact crossover (crossed CIs).

Cohort	Encoder	Training	Crossover [95% CI]
Dermoscopy hair	DINOv2 ViT-S/14	frozen	+0.163 [+0.130, +0.195]
Dermoscopy hair	DINOv2 ViT-B/14	frozen	+0.147 [+0.111, +0.185]
Dermoscopy hair	DINOv2 ViT-L/14	frozen	+0.153 [+0.110, +0.195]
Dermoscopy hair	DermLIP	frozen	+0.014 [−0.037, +0.063]
Thyroid	DINOv2 ViT-S/14	frozen	+0.247 [+0.205, +0.288]
Thyroid	DINOv2 ViT-B/14	frozen	+0.239 [+0.196, +0.280]
Thyroid	DINOv2 ViT-L/14	frozen	+0.263 [+0.218, +0.307]
Thyroid	MedSigLIP-448	frozen	+0.195 [+0.153, +0.238]
Thyroid	ConvNeXt-384	frozen	+0.275 [+0.231, +0.320]
Ovary	DINOv2 ViT-S/14	frozen	+0.296 [+0.208, +0.384]
Ovary	DINOv2 ViT-B/14	frozen	+0.170 [+0.087, +0.252]
Ovary	DINOv2 ViT-L/14	frozen	+0.207 [+0.127, +0.286]
Ovary	MedSigLIP-448	frozen	+0.178 [+0.109, +0.248]
Capsule	DINOv2 ViT-S/14	frozen	+0.370 [+0.325, +0.415]
Capsule	DINOv2 ViT-B/14	frozen	+0.377 [+0.327, +0.426]
Capsule	DINOv2 ViT-L/14	frozen	+0.417 [+0.369, +0.464]
Capsule	MedSigLIP-448	frozen	+0.335 [+0.286, +0.383]
Capsule	ConvNeXt-384	frozen	+0.320 [+0.268, +0.372]
Dermoscopy hair	ResNet-50	fine-tuned	+0.143 [+0.054, +0.232]
Thyroid	ResNet-50	fine-tuned	+0.063 [−0.022, +0.144]
Thyroid	ViT-S	fine-tuned	+0.122 [+0.039, +0.211]
Capsule	ResNet-50	fine-tuned	+0.606 [+0.495, +0.716]
Ovary	ResNet-50 (15 clusters)	fine-tuned	+0.058 [−0.062, +0.179]
Dermoscopy hair, ISIC 2020 (external)	DINOv2 ViT-B/14	frozen, new patients	+0.250 [+0.174, +0.327]

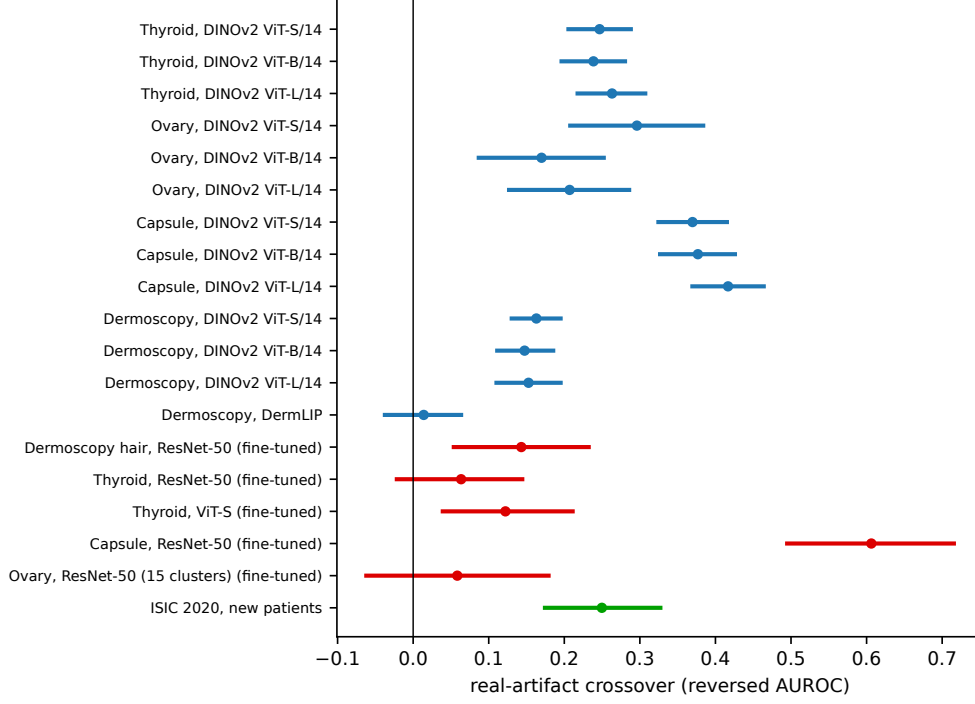


Figure 2: Crossover across encoders, scales, fine-tuned networks (red) and new patients (green); crossed 95% CIs.

5.4 The DermLIP exception

With DermLIP, masking harms at both hair locations (Trap A -0.104 $[-0.143, -0.067]$, Trap B -0.090 $[-0.135, -0.046]$). Two measurements explain it. First, masking costs DermLIP disease signal wherever the hair lies: clean-test AUROC changes by -0.066 $[-0.089, -0.043]$ and -0.063 $[-0.094, -0.033]$. Second, the pixel part of its hair shortcut is small at both locations (oracle hair inpainting $+0.033$ $[+0.025, +0.042]$ and $+0.040$ $[+0.015, +0.064]$), whereas group balancing, which also removes correlates of hair, gains $+0.146$ $[+0.122, +0.170]$ and $+0.110$ $[+0.089, +0.132]$: DermLIP carries the hair shortcut mostly through correlates that masking does not remove at either location. The theory anticipates this from the clean and correlated AUROC alone: predicted crossover $+0.013$, observed $+0.014$.

5.5 The theory, judged by held-out error

Table 7 and Figure 3. Over 136 primary cells the theory predicts the reversed AUROC — never used in fitting — with mean absolute error 0.052 (Pearson r 0.914; 66% within 0.05), against 0.129 for the symmetric-reversal heuristic and 0.237 for “reversed = clean”. It gets the sign of all 9 real-trap crossovers right (MAE 0.015) and of 3/3 fine-tuned crossovers, and the sign of 44 of 50 controlled-sweep gains (heuristic: 38). The account is simple: a model’s reliance on an artifact is fixed by the ratio of artifact to disease signal it sees; masking lowers the disease signal (less context) and leaves an in-ROI artifact’s signal intact, so it raises that ratio exactly when the artifact is inside the ROI.

Table 7: Held-out prediction of the reversed-test AUROC (never used in fitting): the two-parameter theory fitted to each cell’s clean and correlated AUROC, against two parameter-free heuristics.

Cells	Predictor	n	MAE	Pearson r	within 0.05
Primary cells (ERM, mask)	linear-Gaussian theory	136	0.052	0.914	66%
	heuristic: symmetric reversal	136	0.129	0.743	40%
	heuristic: reversed = clean	136	0.237	0.550	18%
All ERM-head arms	linear-Gaussian theory	598	0.036	0.933	79%
	heuristic: symmetric reversal	598	0.076	0.823	65%
	heuristic: reversed = clean	598	0.151	0.584	40%
Fine-tuned (outside assumptions)	linear-Gaussian theory	12	0.027	0.980	92%

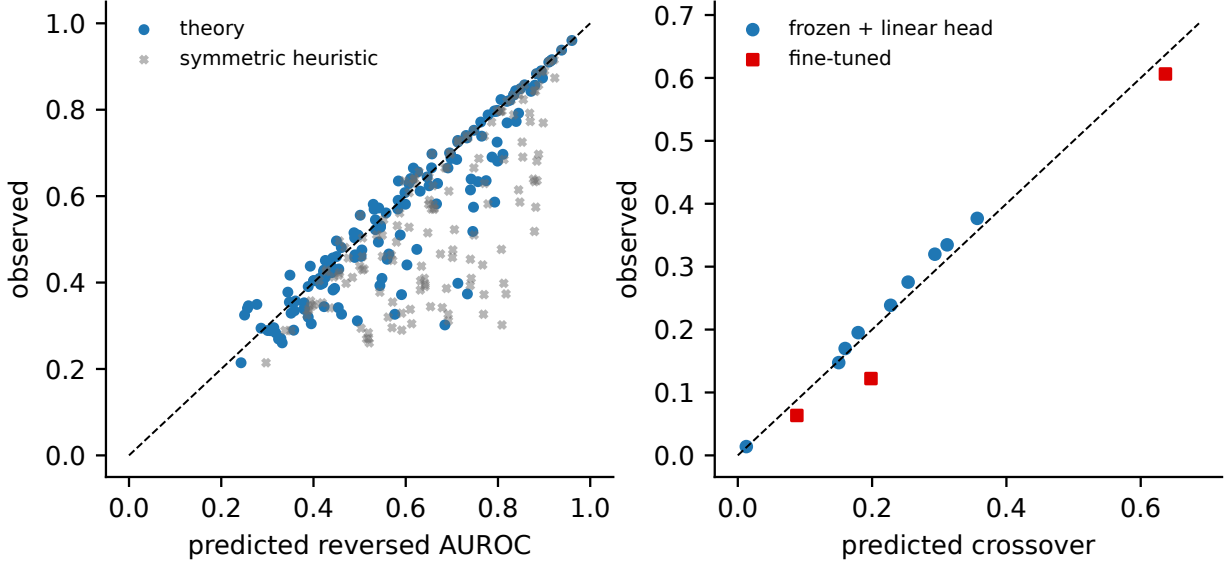


Figure 3: Left: predicted versus observed reversed AUROC (theory, blue; symmetric heuristic, grey). Right: predicted versus observed real-trap crossover (frozen encoders, blue; fine-tuned, red).

6 What failed or changed

- **Natural test sets:** N3 holds in three of five; HAM goes the other way and capsule is inconclusive. With the corrected bootstrap, masking’s gain on easy thyroid pairs and U-MtE’s gain over masking on hard thyroid pairs are no longer significant.
- **The thyroid subgroup** is a post hoc subgroup; its test was registered only after its AUROC result was known. Its interval is now the corrected one.
- **Fine-tuning:** thyroid ResNet-50 is not significant; the archived ovary ResNet-50 run could not be regenerated (no saved predictions) and is withdrawn; the 15-cluster ovary re-run is uninformative.
- **DermLIP:** no crossover; explained, and predicted by the theory, but still an exception.
- **Absolute performance** is below task-specific fine-tuned models on thyroid (frozen DINOv2 with a linear head reaches a clean AUROC of 0.735 on the official split).

7 Conclusion

Supported. On unaltered data masking harms the shortcut-conflicting cases in three of five test sets and lowers thyroid sensitivity at every validation-fixed operating point (registered decision

rule met). The crossover replicates across encoder families and sizes, in three of five fine-tuned networks and on new patients. A two-parameter theory predicts held-out reversed AUROC far better than heuristics and anticipates the DermLIP exception. **Qualified.** The 78-nodule subgroup is exploratory confirmation. **Not supported.** Harm on natural HAM and capsule test sets; the crossover for DermLIP and for two fine-tuned networks.

8 Questions for the supervisor

1. Is a sensitivity loss on one patient-disjoint split (thyroid) enough for the clinical claim in the abstract, or should the abstract lead with the AUROC on shortcut-conflicting cases?
2. Should the theory be a main contribution (with its held-out error as evidence) or a supporting section?
3. Is ISIC 2020 enough external validation, given that it is the same modality and artifact as the main dermoscopy analysis?

9 Next stage

Stage 6 turns to what to do instead of masking (the decision guide, with its remedies tested in the primary family), the blinded image audit prepared for a clinician, the limitations, the paper itself, the CLAIM checklist and the submission statements.

References

Haifan Gong, Hui Cheng, Yifan Xie, Shuangyi Tan, Guanqi Chen, Fei Chen, and Guanbin Li. Less is more: Adaptive curriculum learning for thyroid nodule diagnosis. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, volume 13434 of *LNCS*, 2022. doi: 10.1007/978-3-031-16440-8_24.